Innovation and its type

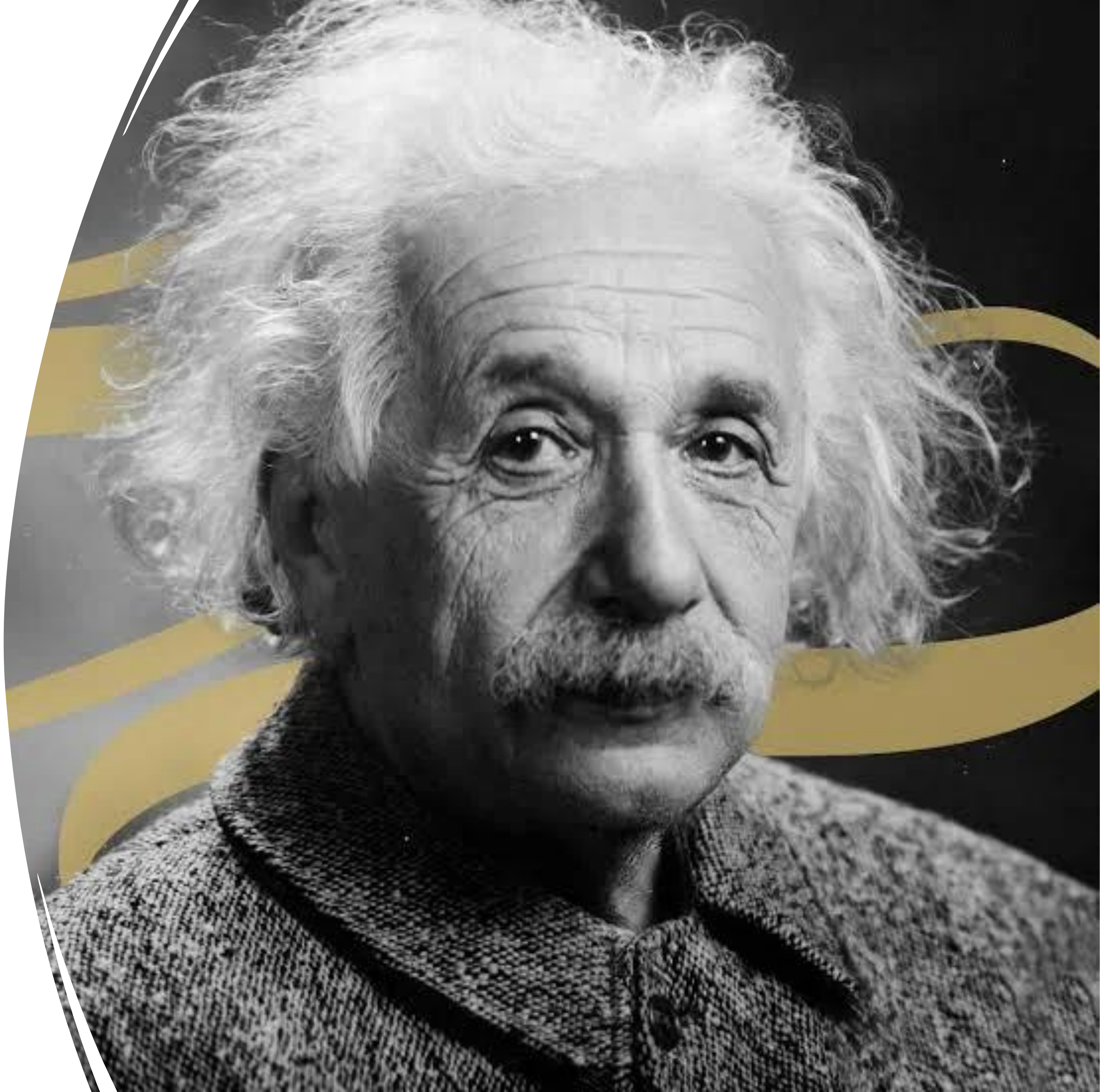
Week 5

Examining the Future Trajectory of Entrepreneurship

20 mins Class activity

- You can’t solve a problem on the same level that it was created. You have to rise above it to the next level.

Albert Einstein



Tasks to do now (15 mins maximum)

Arrange yourselves into groups of three to four Take a blank page write the names on it.

First identify a problem. In detail describe the problem. Make sure the explanation is clear and precise as possible

In the same groups formulate a short solution for this problem. In details describe the solution.

1. Redefining Entrepreneurship in Technology

- The authors argue that entrepreneurship is more than just starting a business—it's about creating value through innovation and opportunity recognition.
- In the context of computer science, how can you apply this broader definition of entrepreneurship when developing software, apps, or AI solutions?

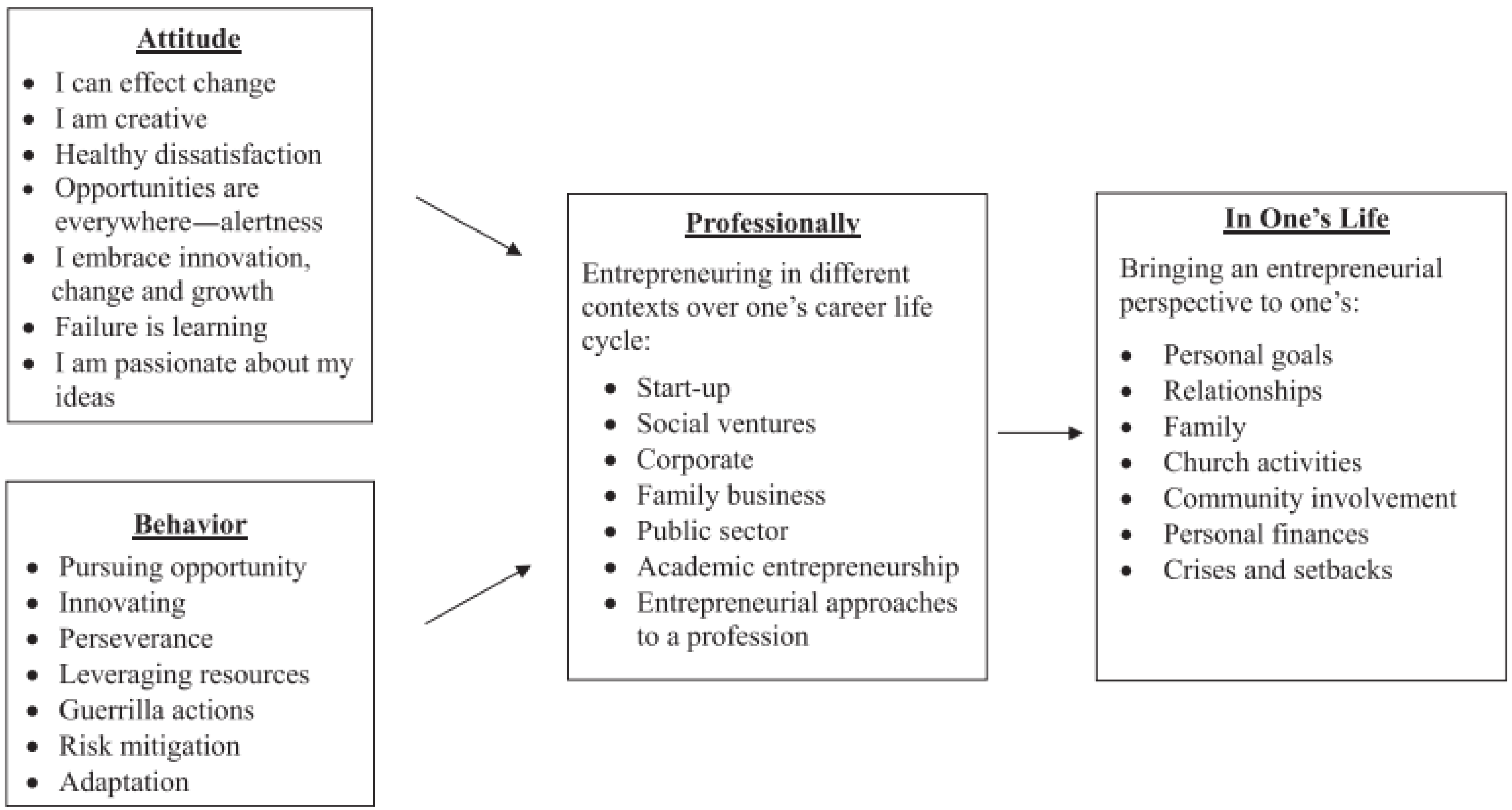
- Tech entrepreneurship is not limited to launching a company. It includes creating new platforms, software tools, or systems that generate value. For example, developing a health-monitoring AI or an educational app for rural schools embodies entrepreneurship. Students can think of innovation as solving user problems through code or technology.

2. Learning by Doing in Technology Entrepreneurship

1. Kuratko and Morris emphasize experiential learning such as real projects and incubator experiences
2. What kind of hands-on learning experiences can help you as computer science students think and act more entrepreneurially?

- Experiential learning could include hackathons, prototype building, internships in startups, or launching a small tech-based venture. Such activities allow students to test their coding, teamwork, and design skills in real settings where they must meet customer or user needs.

Figure 1
A Framework for Teaching the Entrepreneurial Mindset

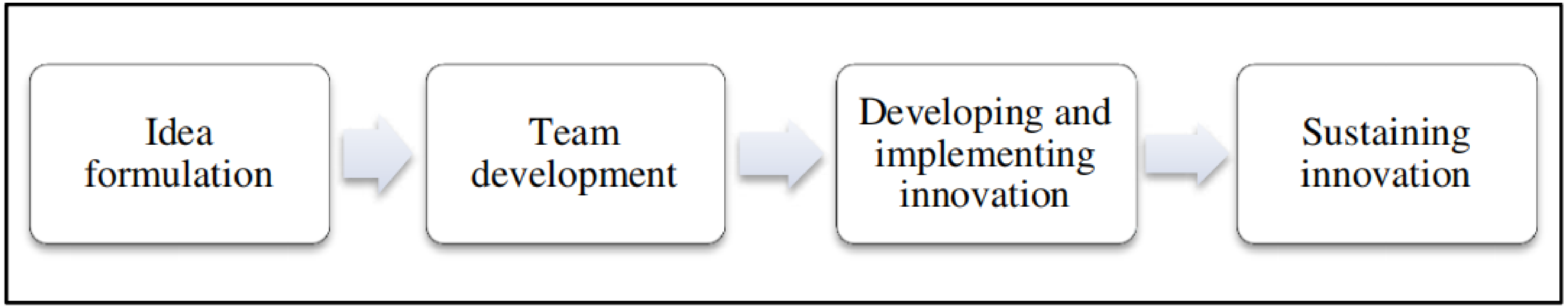


INNOVATION

Innovation is defined as the process of bringing about new ideas, methods, products, services, or solutions that have a significant positive impact and value.

INNOVATION PROCESS

A **change** process that is deliberately undertaken to realise a **novel** product, process or practice in order to **improve** outcomes.



INVENTION VS INNOVATION VS CREATIVITY

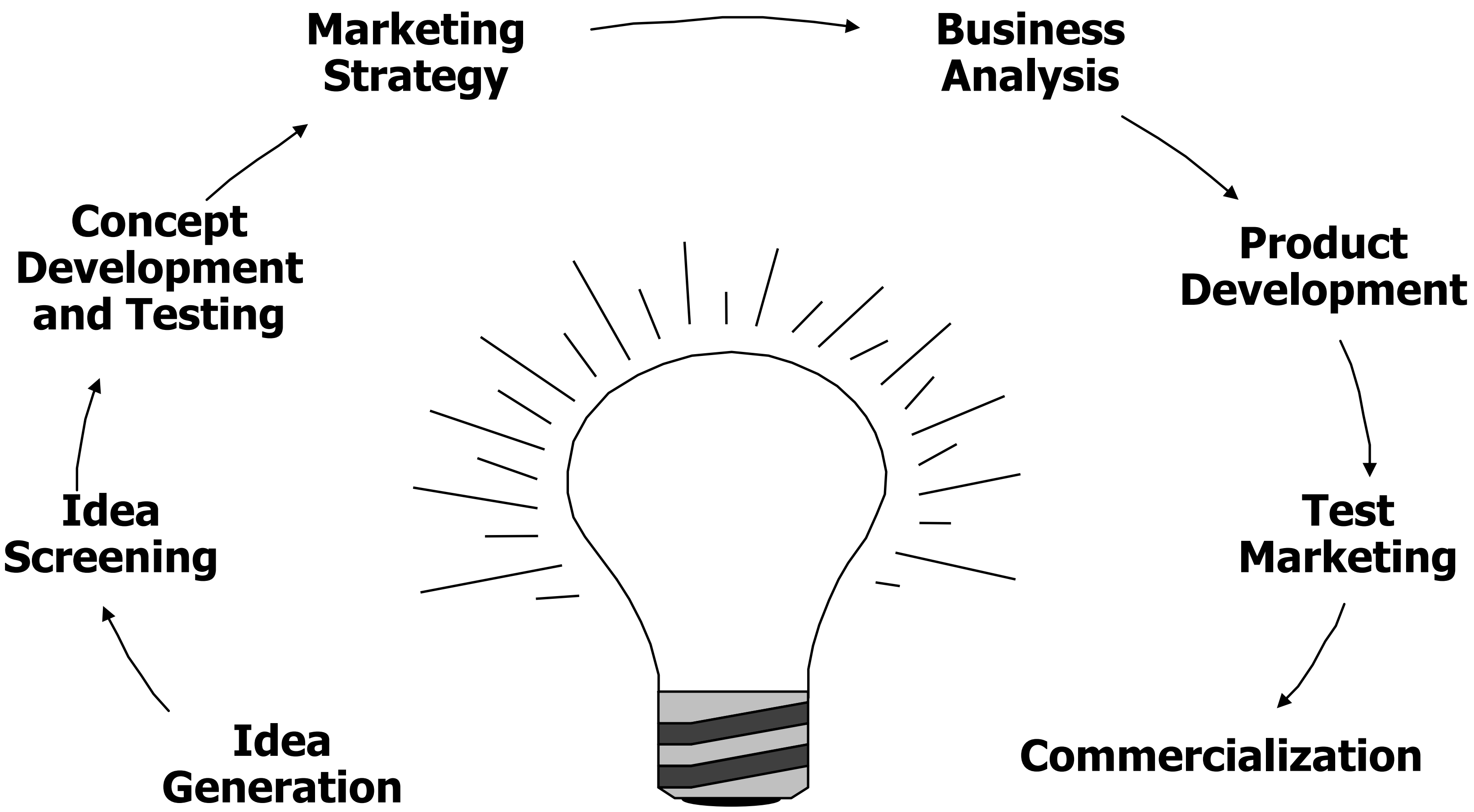
- **The invention** is the process of creating something new that did not previously exist. It can be a physical object, a process, a system, or an idea.
- Invention often involves taking existing ideas and combining them in new ways or using existing materials to create something novel.
- The invention of the telephone, for example, involved combining the idea of the telegraph with the ability to transmit sound.
- New, Value, Real Life

- **Innovation,** on the other hand, involves taking something that already exists and improving or modifying it in some way.
- Innovation can involve making a product more efficient, improving its functionality, or making it more user-friendly.
- The introduction of touchscreens on smartphones, for example, was an innovation that made interacting with mobile devices easier and more intuitive.

- **Creativity** is the ability to generate new ideas or concepts. It involves thinking outside the box and seeing things in a new light.
- Creativity can be expressed in many different ways, including art, music, writing, and problem-solving.
- Creativity is an important aspect of both invention and innovation. Without creativity, it would be difficult to come up with new ideas and solutions.

Invention precedes innovation through creativity

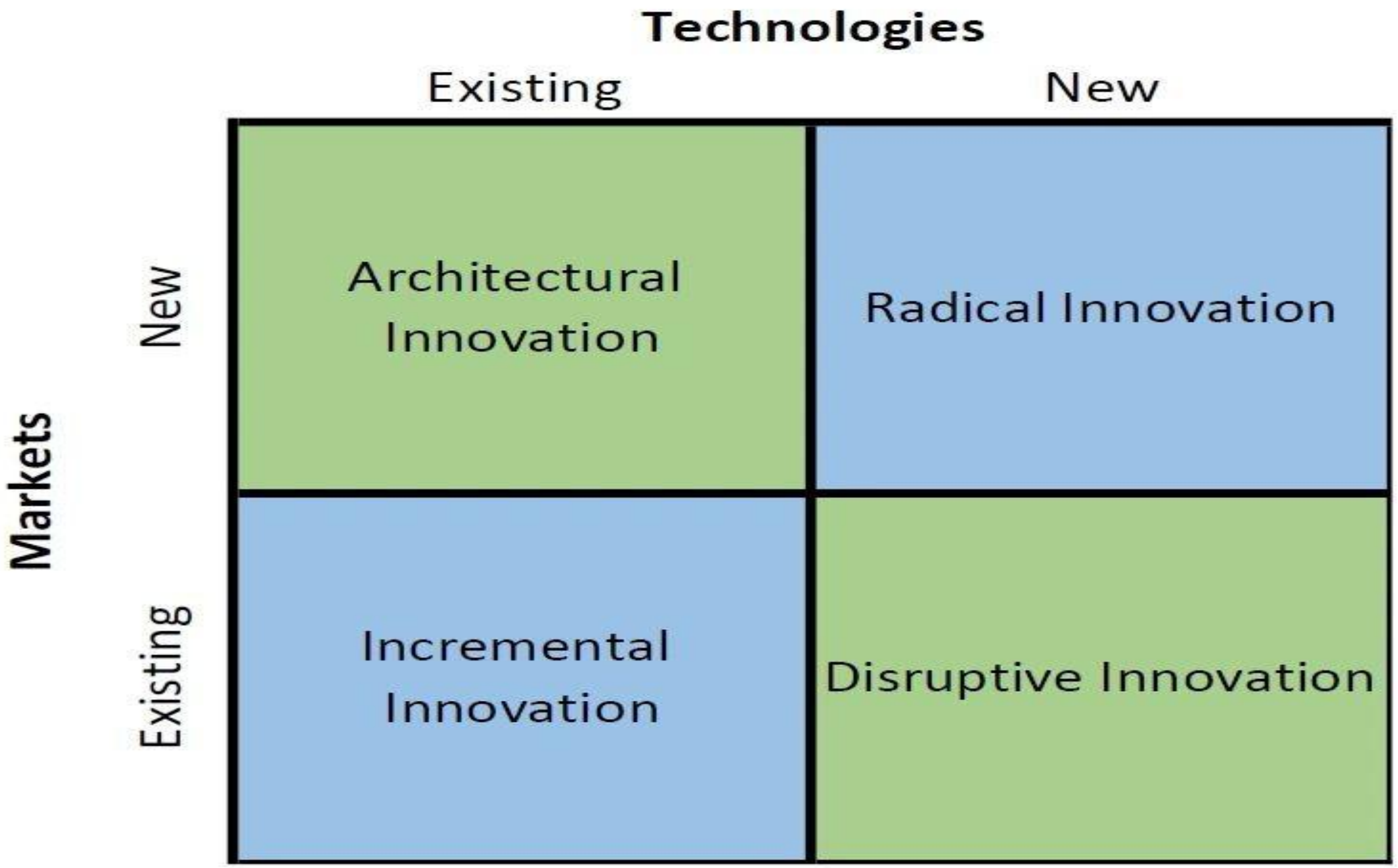
MAJOR STAGES IN NEW-PRODUCT DEVELOPMENT



INNOVATION MATRIX

- Innovation can be a confusing topic because there are so many different kinds of innovations out there and everyone seems to use the term differently.
- An innovation matrix lets organizations manage and prioritize ideas via visualization tools. It enables companies of all sizes to collect ideas and decide which ones are worth pursuing, depending on company or team goals.
    - Optimize idea generation
    - Improve idea management
    - Prioritize ideas for better decision-making
    - Targeted innovation
- **It can be classified based on two criteria**
    - **Markets**
    - **Technology**

INNOVATION TAXONOMY



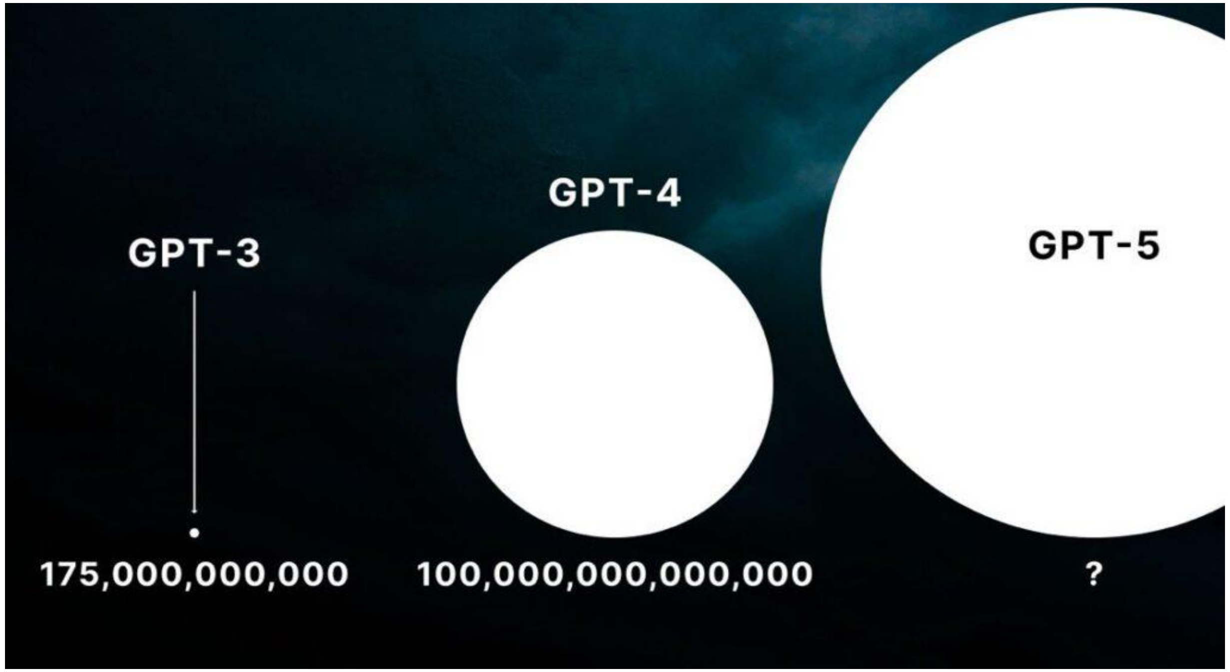
INCREMENTAL INNOVATION

- ✓ Incremental Innovation is the most common form of innovation.
- ✓ Small improvements to existing products, services, or processes targeting familiar markets with well-understood technology.
- ✓ It utilizes your existing technology and increases value to the customer (features, design changes, etc.) within your existing market.
- ✓ Almost all companies engage in incremental innovation in one form or another.


HONDA
The Power of Dreams
HONDA
環境負荷ゼロの循環型社会を目指す
Honda strives for a "circular / resource-circulation society"
that aims for "zero environmental impact"


IPHONE 12
A14 Bionic Chip
4GB RAM
64GB/128GB/256GB
IPHONE 13
A15 Bionic Chip
4GB RAM
128GB/256GB/512GB
IPHONE 14
A15 Bionic Chip
6GB RAM
128GB/256GB/512GB


Windows 7/8.1/10/11 All in One ISO
Windows 7
Windows 8
Compatible
Windows 10
Windows 11


GPT-3
GPT-4
GPT-5
175,000,000,000
100,000,000,000,000
?

DISRUPTIVE INNOVATION

- New technology is applied to an existing market, offering a new value proposition and potentially overtaking established market leaders over time.
- Disruptive innovation, also known as stealth innovation, involves applying new technology or processes to a current market.


iPhone
June 29, 2007
"Apple reinvents the phone."
Thickness
11.6 mm
Weight
135 g (4.8 oz)
Display
3.5"
Storage
4, 8, 16 GB
Camera
2.0 MP
Memory
128 MB
Multitouch
Proximity sensor
Ambient light sensor
Accelerometer
~300,000 sold in first weekend


Kodak
CHAPTER 11
Fuji
SONY
Canon
SAMSUNG
DAVE GRANLUND © www.davegranlund.com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Nikon

ARCHITECTURAL INNOVATION

- Existing technologies applied to new markets involve reconfiguring product or system components for new applications or uses.
- Architectural innovation is simply taking the lessons, skills and overall technology and applying them within a different market.
- This innovation is amazing at increasing new customers as long as the new market is receptive.
- Most of the time, the risk involved in architectural innovation is low due to the reliance and reintroduction of proven technology.


3M Science.
Applied to Life.
Scotch-Brite
Premium Kitchen Towel
3M
MICROFIBER DETAIL CLOTH
Stainless Steel Scrub
Sponge Wipe
S-shape
Extra Power
Scotch-Brite
Heavy Duty
Fights Tough Stains
Scrub Pad
Stays Firm
Gentle Care


amazon


3M Lava™
Crowns and Bridges


amazon books

How can formula one change the face of the airline industry?

RADICAL INNOVATION

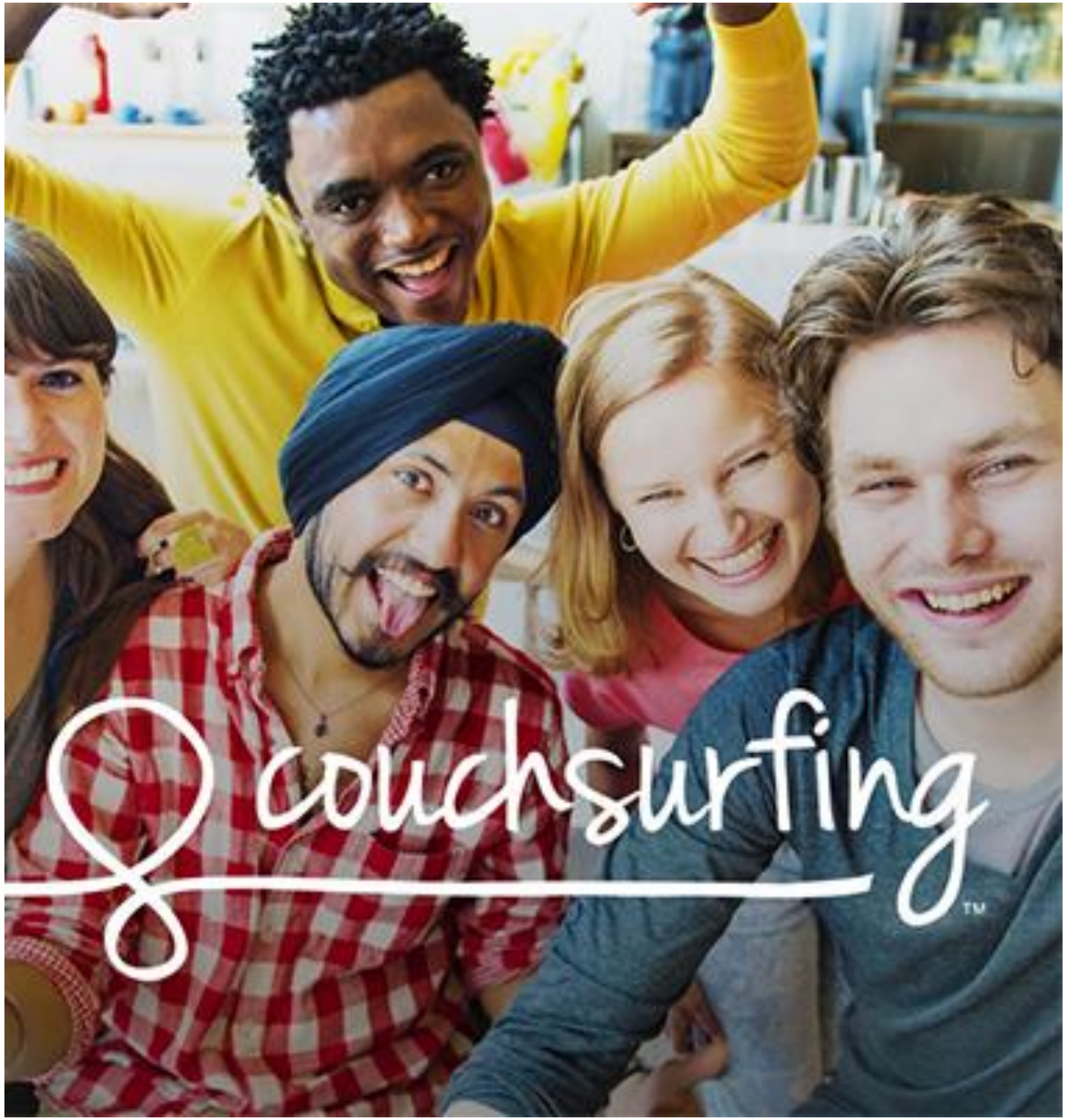
Radical innovation is what we think of mostly when considering innovation.

It gives birth to new industries (or swallows existing ones) and involves creating revolutionary technology.

The airplane, for example, was not the first mode of transportation, but it is revolutionary as it allowed commercialized air travel to develop and prosper.

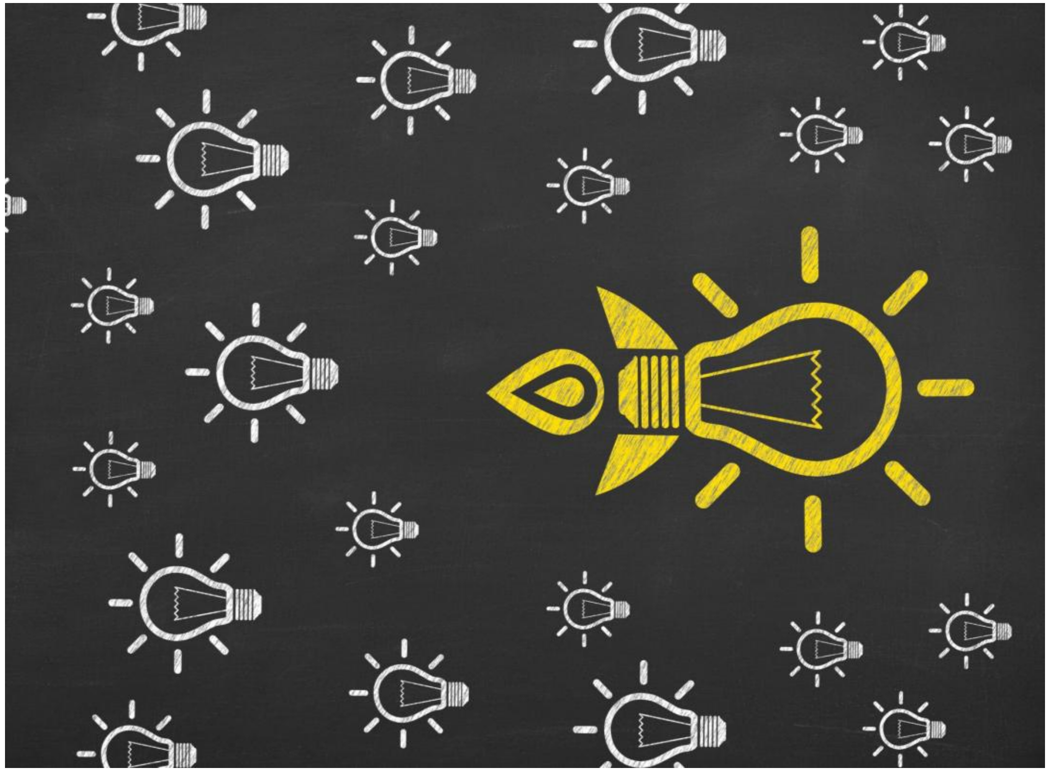

CHAT GPT


BITCOIN
PEER TO PEER


couchsurfing

SOME OTHER TYPES OF INNOVATION

- Product Innovation (Apple)
- Service innovation (Emirates)
- Process Innovation (Toyota Just in time)
- Business Model Innovation (Netflix)
- Organizational Innovation (google)
- Marketing Innovation (coca cola)



CLASS ACTIVITY

Imagine you are leading a team tasked with developing a new product or technology for a well-established company in a traditional industry, like manufacturing. Your goal is to introduce innovation into the company's product line.

1. Identify the type(s) of innovation (incremental, disruptive, radical, or architectural) that would best suit your strategy.
2. Provide a real-world example of a company that successfully combined innovation with marketing strategies in a traditional industry. Explain their approach and the outcomes achieved."